Orbit SaaS - Mobile Conversion Analysis

Comprehensive Data Analytics Report

Executive Summary

This report presents a comprehensive analysis of Orbit SaaS's mobile application conversion challenges. Through systematic data analysis of 2,000 users across 24 months (2023-2024), we identified critical technical and user experience issues causing 48% of mobile users to abandon the application within the first two minutes.

Key Findings:

- Android users generate 2x more support tickets than iOS users
- iOS Chrome compatibility issues result in only 6.06% signup rate
- Basic plan mobile churn rate of 24.39% represents ~$45,000 annual revenue loss
- 48% of churn occurs at first impression stage

Business Impact: Implementing recommended technical fixes and UX improvements can recover up to 50% of lost revenue and significantly improve mobile platform performance within 6 months.

Table of Contents

1. Introduction & Business Context
2. Data Infrastructure & Methodology
3. Analysis Findings
4. Strategic Recommendations
5. Implementation Roadmap
6. Appendix: SQL Queries

---

1. Introduction & Business Context

1.1 Business Problem Statement

Orbit, a cloud-based project management and team collaboration SaaS platform, experienced rapid growth in its first year but encountered a significant slowdown in the past six months. The primary concern identified by CEO Elif was the substantial gap in conversion rates between web and mobile platforms.

Critical Business Questions:

1. Why is mobile application conversion rate significantly lower than web?
2. What factors are causing the recent growth slowdown?
3. At which stage of the user journey are customers being lost?

1.2 Data Sources

The analysis integrated three primary data sources:

- **Google Analytics 4 (GA4):** User behavior and event tracking (23,262 events)
- **CRM System:** Customer subscription and support data (2,000 users)
- **Advertising Platforms:** Campaign performance data from Google Ads and Meta

---

2. Data Infrastructure & Methodology

2.1 AWS Architecture

The project implemented a modern data lakehouse architecture on AWS:

Data Pipeline:

Airbyte → Amazon S3 → AWS Glue → Amazon Athena → Amazon QuickSight

Infrastructure Components:

- **Data Ingestion:** Airbyte scheduled weekly data extraction
- **Data Lake:** Amazon S3 with partitioned storage structure
- **ETL Processing:** AWS Glue for data transformation and cleaning
- **Analytics:** Amazon Athena for SQL-based analysis
- **Visualization:** Amazon QuickSight for executive dashboards
- **Monitoring:** CloudWatch alarms for pipeline health

2.2 Data Partitioning Strategy

Table	Partition Key	Purpose
ga4_events	event_date	Time-based query optimization
users	created_date	Cohort analysis efficiency
crm_subscriptions	start_date	Subscription trend analysis
ads_campaign_performance	date	Campaign performance tracking

2.3 Analysis Methodology

Analytical Approach:

1. Platform comparison analysis (iOS vs Android vs Web)
2. Conversion funnel breakdown by device type
3. Churn pattern identification
4. Channel quality assessment
5. Sector-specific performance evaluation

---

3. Analysis Findings

3.1 Technical Infrastructure Issues

3.1.1 Platform Support Ticket Distribution

Finding: Android users generated twice the volume of support tickets compared to iOS users.

Platform	Total Tickets	Technical Issues	Billing Issues
Android	1,568	608	348
iOS	783	318	173

Analysis: Android users experience 2x more support tickets overall, with technical issues being the dominant category (608 vs 318). This indicates significant technical debt in the Android application that directly impacts user experience and conversion rates.

3.1.2 Device/Browser Performance Analysis

Finding: iOS Chrome users face severe signup difficulties due to compatibility issues.

Platform	Browser	Unique Users	Signups	Signup Rate
iOS	Chrome	924	56	6.06%
iOS	Safari	732	48	6.56%
Android	Safari	1,210	102	8.43%
Android	Chrome	1,412	155	10.98%

Analysis: The iOS Chrome combination shows critically low signup performance at 6.06%, nearly half the rate of Android Chrome (10.98%). This suggests WebKit restrictions on iOS are causing form handling and compatibility issues that prevent successful account creation.

3.2 Customer Churn Root Causes

3.2.1 Churn Driver Analysis

Finding: Technical issues are the primary driver of mobile customer churn.

Churn Reason	Count	Percentage
Technical Issues	66	41%
Feature Requests	34	21%
Billing Issues	33	20%
Bug Reports	29	18%

Analysis: Technical problems account for 41% of all mobile churn, representing the single largest controllable factor. This aligns with the high support ticket volume observed in platform analysis.

3.2.2 Churn Moment Analysis

Finding: 48% of churn occurs during the first impression stage.

User Action	Churn Count	Cumulative %
Session Start	141	24%
Page View	136	48%
Login	89	63%
Feature Used	65	75%

Analysis: The concentration of churn at session start and page view stages (277 out of 568 total, 48%) indicates fundamental onboarding and first-impression experience failures. Users are abandoning the application before engaging with core features.

3.3 Subscription Plan Analysis

Finding: Higher-tier plans show dramatically better retention rates.

Plan Type	Total Users	Churned Users	Churn Rate
Basic	529	129	24.39%
Pro	275	42	15.27%
Enterprise	71	1	1.41%

Financial Impact:

- Basic plan churn: 129 × $348 ARR = $44,892 annual loss
- Pro plan churn: 42 × $948 ARR = $39,816 annual loss
- Enterprise plan churn: 1 × $2,388 ARR = $2,388 annual loss

Analysis: The inverse relationship between plan price and churn rate (24.39% → 15.27% → 1.41%) demonstrates that committed, high-value customers are more tolerant of mobile UX issues. This suggests opportunity for significant revenue recovery by improving Basic plan user experience.

3.4 Marketing Channel Quality Assessment

Finding: Referral and email channels show unexpectedly high churn rates.

Channel	Total Users	Churned	Churn Rate
Referral	262	90	34.35%
Email	241	82	34.02%
Content Marketing	270	88	32.59%
Social Media	245	77	31.43%
Paid Search	269	81	30.11%
Organic Search	283	78	27.56%
Direct	277	72	25.99%

Analysis: Channels traditionally considered "high quality" (Referral, Email) are producing the highest churn rates. This counter-intuitive finding suggests:

1. Referral program may lack quality controls
2. Email lists require segmentation and cleaning
3. Direct traffic users show highest intent and best retention

3.5 Sector Performance Differences

Finding: Traditional industries struggle with mobile adoption.

Industry	Total Users	Churned	Churn Rate
Manufacturing	245	83	33.88%
Healthcare	243	82	33.74%
Consulting	224	71	31.70%
Marketing	247	77	31.17%
Finance	196	60	30.61%
Education	214	64	29.91%

Industry	Total Users	Churned	Churn Rate
Technology	233	64	27.47%
Retail	245	67	27.35%

Analysis: Traditional sectors (Manufacturing, Healthcare) show 6+ percentage point higher churn than tech-savvy industries (Technology, Retail). This indicates that mobile UX expectations vary significantly by industry vertical, requiring sector-specific design considerations.

3.6 Page and Feature Analysis

3.6.1 Most Problematic Pages

Page	Total Visitors	Churned Users	Churn Rate
/projects	330	89	26.97%
/team	316	85	26.90%
/billing	329	82	24.92%
/settings	330	80	24.24%
/reports	343	83	24.20%
/dashboard	319	71	22.26%

Analysis: Core functionality pages (/projects, /team) show highest churn rates, while the dashboard overview page performs best. This indicates complex features are not mobile-optimized, causing user frustration at the point of actual work execution.

3.6.2 Most Problematic Features

Feature	Churned Users
Project Creation	27
Push Notifications	26
Task Management	26
Calendar Integration	26
Reporting	23

Analysis: Project Creation leads churn triggers, consistent with page analysis showing /projects as most problematic. Push Notifications appearing as second-highest suggests notification strategy may be irritating users rather than engaging them.

---

4. Strategic Recommendations

4.1 Immediate Action Items (Month 1)

Priority 1: iOS Chrome Compatibility Resolution

Problem: 6.06% signup rate on iOS Chrome **Target:** Increase to 15% within 30 days **Action Plan:**

- Conduct comprehensive WebKit compatibility audit
- Implement form validation workarounds for iOS constraints
- Add fallback mechanisms for unsupported browser features
- Deploy A/B testing to validate improvements

Expected Impact: +150% signup rate improvement, ~200 additional monthly signups

Priority 2: Mobile Onboarding Redesign

Problem: 48% churn at first impression **Target:** Reduce to 25% within 60 days **Action Plan:**

- Simplify initial screens to 3-step onboarding flow
- Implement progressive disclosure of features
- Add interactive tutorials for core functionality
- Create mobile-specific welcome experience

Expected Impact: -48% first-impression churn, +23% net mobile user retention

Priority 3: Android Technical Stability

Problem: 2x support ticket volume vs iOS **Target:** Reduce Android tickets by 50% within 90 days **Action Plan:**

- Establish dedicated Android QA testing cycle
- Implement automated crash reporting and monitoring
- Prioritize top 10 technical issues from support tickets
- Deploy staged rollout strategy for Android releases

Expected Impact: Improved user satisfaction, reduced support costs

4.2 Mid-Term Improvements (Months 2-6)

Initiative 1: Mobile-First Feature Development

Focus Areas:

- Project creation mobile workflow simplification
- Touch-optimized team management interface

- Mobile-friendly billing and subscription management

Initiative 2: Sector-Specific Customization

Target Sectors: Manufacturing, Healthcare (33%+ churn rate) **Approach:**

- Develop desktop-like mobile interface option
- Create industry-specific onboarding flows
- Implement role-based UI adaptation

Initiative 3: Marketing Channel Optimization

Focus: Referral (34.35%) and Email (34.02%) channels **Actions:**

- Implement referral quality scoring
- Create email list segmentation by engagement
- Develop channel-specific landing pages

4.3 Performance Targets

Metric	Current	3-Month Target	6-Month Target
iOS Chrome Signup Rate	6.06%	12%	15%
First Impression Churn	48%	35%	25%
Android Support Tickets	1,568/year	1,100/year	780/year
Basic Plan Churn Rate	24.39%	20%	15%
Overall Mobile Conversion	Baseline	+25%	+50%

4.4 Expected Financial Impact

Revenue Recovery Calculation:

- Current Basic plan loss: $44,892/year
- 50% churn reduction: $22,446 recovered revenue
- Improved conversion rate (+25%): $30,000 additional ARR
- **Total potential impact: $52,446 annual recurring revenue**

---

5. Implementation Roadmap

Month 1: Foundation

- Week 1-2: iOS Chrome compatibility fixes

- Week 3-4: Mobile onboarding redesign and testing

Month 2: Stabilization

- Week 5-6: Android technical debt reduction sprint
- Week 7-8: Initial deployment and monitoring

Month 3: Optimization

- Week 9-10: Feature-level mobile improvements
- Week 11-12: Channel quality improvements

Months 4-6: Scaling

- Sector-specific customizations
- Advanced analytics implementation
- Continuous improvement cycle

Success Metrics Dashboard

Weekly tracking of:

- Platform-specific signup rates
- Conversion funnel drop-off points
- Support ticket volume by category
- Churn rate by plan and channel

---

6. Conclusion

The mobile conversion challenge at Orbit SaaS is multi-dimensional, stemming from technical infrastructure issues, user experience gaps, and channel quality problems. However, the analysis reveals clear, actionable opportunities for improvement with quantifiable business impact.

The concentration of issues in specific, addressable areas (iOS Chrome compatibility, first-impression experience, Android stability) means that focused engineering effort can yield significant returns. The strong retention performance of Enterprise users demonstrates that the core product value proposition is sound—the challenge lies in delivering that value effectively through mobile platforms.

By implementing the recommended immediate actions and following the phased roadmap, Orbit can realistically achieve 50% mobile conversion improvement within six months, translating to over $50,000 in recovered annual recurring revenue.

Appendix A: Sample SQL Queries

Query 1: Platform Churn Rate Analysis

```
WITH plan_totals AS (
  SELECT
    cs.plan_type,
    COUNT(DISTINCT cs.user_id) as total_users
  FROM "orbit_analytics"."crm_subscriptions" cs
  JOIN "orbit_analytics"."ga4_events" ge ON cs.user_id = ge.user_id
  WHERE ge.device_category = 'Mobile'
  GROUP BY cs.plan_type
),
plan_churn AS (
  SELECT
    cs.plan_type,
    COUNT(DISTINCT u.user_id) AS churned_count
  FROM "orbit_analytics"."ga4_events" ge
  JOIN "orbit_analytics"."crm_subscriptions" cs ON ge.user_id = cs.user_id
  JOIN "orbit_analytics"."users" u ON cs.user_id = u.user_id
  WHERE ge.device_category = 'Mobile'
  AND u.is_churned = true
  GROUP BY cs.plan_type
)
SELECT
  pc.plan_type,
  pc.churned_count,
  pt.total_users,
  ROUND(
    CAST(pc.churned_count AS DOUBLE) / CAST(pt.total_users AS DOUBLE) * 100,
    2
  ) AS churn_rate_percent
```

```
FROM plan_churn pc
JOIN plan_totals pt ON pc.plan_type = pt.plan_type
ORDER BY churn_rate_percent DESC;
```

Query 2: Page-Level Churn Analysis

```
WITH page_totals AS (
  SELECT
    regexp_extract(ge.event_params, 'page_location":\s*"([^"]+)"', 1) AS page_location,
    COUNT(DISTINCT ge.user_id) as total_visitors
  FROM "orbit_analytics"."ga4_events" ge
  WHERE ge.device_category = 'Mobile'
  AND ge.event_params LIKE '%page_location%'
  GROUP BY regexp_extract(ge.event_params, 'page_location":\s*"([^"]+)"', 1)
),
churned_pages AS (
  SELECT
    regexp_extract(ge.event_params, 'page_location":\s*"([^"]+)"', 1) AS page_location,
    COUNT(DISTINCT ge.user_id) AS churned_user_count
  FROM "orbit_analytics"."ga4_events" ge
  JOIN "orbit_analytics"."users" u ON u.user_id = ge.user_id
  WHERE u.is_churned = true
  AND ge.device_category = 'Mobile'
  AND ge.event_params LIKE '%page_location%'
  GROUP BY regexp_extract(ge.event_params, 'page_location":\s*"([^"]+)"', 1)
)
SELECT
  cp.page_location,
  cp.churned_user_count,
  pt.total_visitors,
  ROUND(
```

```
        CAST(cp.churned_user_count AS DOUBLE) / CAST(pt.total_visitors AS DOUBLE) * 100,
        2
    ) AS page_churn_rate_percent
FROM churned_pages cp
JOIN page_totals pt ON cp.page_location = pt.page_location
ORDER BY page_churn_rate_percent DESC;
```

Query 3: Device/Browser Performance

```
SELECT
    operating_system,
    browser,
    COUNT(DISTINCT user_id) as unique_users,
    COUNT(DISTINCT CASE WHEN event_name = 'signup' THEN user_id END) as signups,
    ROUND(
        CAST(COUNT(DISTINCT CASE WHEN event_name = 'signup' THEN user_id END) AS DOUBLE) * 100.0
/
        CAST(COUNT(DISTINCT user_id) AS DOUBLE), 2
    ) as signup_rate
FROM "orbit_analytics"."ga4_events"
WHERE device_category = 'Mobile'
GROUP BY operating_system, browser
HAVING COUNT(DISTINCT user_id) > 50
ORDER BY signup_rate ASC;
```

Report Prepared By: Data Analytics Team
Date: January 2025
Project: Orbit SaaS Mobile Conversion Analysis
Data Period: January 2023 - December 2024